# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: COMPUTER NETWORKS**
**COURSE CODE: CSA07**

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**COURSE OUTCOME COVERED**

**CO2:** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

*Assessment Tool Weightage: 30%*
*Software:Cisco Packet Tracer,Wireshark*

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**QUESTIONS: 5** 75% 48 **Total Marks:50**

Annexure A

## Debugging & Optimisation Task

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***Code of Conduct:***

*I* J.Manoj (192579143) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** J.Manoj

| | A | B | C | D | E | F | G | H | |
|---|---|---|---|---|---|---|---|---|---|
| Q1 | 2 | 2 | 1 | | | | | | |
| Q2 | 2 | 2 | 1 | | | | | | |
| Q3 | 1.5 | 2.5 | 0.5 | | | | | | |
| Q4 | 1.5 | 1.5 | 1 | | | | | | |
| Q5 | 1 | 1.5 | 1 | | | | | | |

| | Understanding scenario (20%) | identification of key issues (20%) | Application of Concept (25% | Decision making (20%) | Clarity of Response (15%) | |
|---|---|---|---|---|---|---|
| Q1 | 1.5 | 2 | 1.5 | 1.5 | 1.5 | 8 |
| Q2 | 1 | 2.5 | 2 | 1 | 1 | 7.5 |
| Q3 | 2 | 1.5 | 2 | 2 | 1 | 8.5 |
| Q4 | 1.5 | 2 | 1.5 | 1 | 1.5 | 7.5 |
| Q5 | 1 | 1.5 | 1 | 1.5 | 1 | 6 |

37.5

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Annexure A

## Debugging & Optimisation Task

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*I __j.manoj(192572143)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
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***Signature of the student:***
j.manoj

# QUESTION 1:

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

<u>Task</u>

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

## 1. Aim

To analyze the given workstation network parameters, identify misconfiguration errors preventing inter-subnet/intra-subnet communication, perform step-by-step subnetting calculations, and provide an optimized, fully functional IP configuration.

## 2. Given Data

- Workstation IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (Prefix: /26)

- Configured Default Gateway: 192.168.10.1
- Target Subnet: 192.168.10.0/26

## 3. Relevant Formulas & Mathematical Foundations

- Subnet Block Size = 256 − (Decimal Value of Interesting Octet)
- Total Host IP Addresses per Subnet = 2^h (where h is the number of host bits = 32 − Prefix Length)
- Total Usable Hosts = 2^h − 2 (subtracting 1 for Network ID and 1 for Broadcast ID)
- Network Address = (IP Address) AND (Subnet Mask)
- Broadcast Address = Network Address + Block Size − 1

## 4. Step-by-Step Calculations & Technical Analysis

### Step 4.1: Subnet Mask & Block Size Evaluation

The prefix length is /26. In binary representation, the mask is:

```
11111111.11111111.11111111.11000000 = 255.255.255.192
```

The number of network bits n = 26, and the number of host bits h = 32 − 26 = 6.

Calculating block size: Block Size = 256 − 192 = 64 (or 2^6 = 64).

### Step 4.2: Identification of Subnet Ranges

With a block size of 64, the 192.168.10.0 classful network is divided into 4 subnets:

| Subnet | Network ID | Host Range | Broadcast |
|---|---|---|---|
| Subnet 0 | 192.168.10.0 | 192.168.10.1 – 192.168.10.62 | 192.168.10.63 |
| Subnet 1 | 192.168.10.64 | 192.168.10.65 – 192.168.10.126 | 192.168.10.127 |
| Subnet 2 | 192.168.10.128 | 192.168.10.129 – 192.168.10.190 | 192.168.10.191 |
| Subnet 3 | 192.168.10.192 | 192.168.10.193 – 192.168.10.254 | 192.168.10.255 |

### Step 4.3: Verification of Configured Address & Default Gateway

**Configuration Error Identified:**

- Subnet Mismatch: The target network is 192.168.10.0/26 (Subnet 0). However, the assigned IP 192.168.10.65 falls into Subnet 1 (192.168.10.64/26).
- Gateway Mismatch: The configured gateway is 192.168.10.1, which resides in Subnet 0. A workstation in Subnet 1 cannot communicate with a gateway in Subnet 0 directly without local routing, as they belong to separate layer-3 broadcast domains.

## 5. Subnetting Diagram & Network Topology

```
NETWORK TOPOLOGY & SUBNET BOUNDARY DIAGRAM (192.168.10.0/24 divided into /26)

Target Subnet 0 (192.168.10.0/26)        Subnet 1 (192.168.10.64/26)
[Range: 192.168.10.1 to .62]             [Range: 192.168.10.65 to .126]
+-------------------------------+         +-------------------------------+
| Router Interface (Gateway)    |         | MISCONFIGURED WORKSTATION     |
| IP: 192.168.10.1/26           |         | IP: 192.168.10.65/26          |
| (Belongs to Subnet 0)         |         | (Belongs to Subnet 1)         |
+-------------------------------+         +-------------------------------+
          |                                 |
          v                                 v
 [ Cannot communicate across subnet boundaries without local gateway ] <-- FAULT

CORRECTED OPTIMIZED TOPOLOGY:
+---------------------------------------------------------------------+
| Router/Gateway: 192.168.10.1/26 <---> Workstation: 192.168.10.2/26    |
| (Subnet 192.168.10.0/26)                                            |
+---------------------------------------------------------------------+
```

## 6. Optimized Parameters & Summary Table

To place the workstation into the target subnet 192.168.10.0/26 alongside its gateway (192.168.10.1), we assign a valid host IP from Subnet 0 (e.g., 192.168.10.2).

| Parameter | Misconfigured Value | Optimized Value (Corrected) |
|---|---|---|
| Network Address | 192.168.10.64 (for IP .65) | 192.168.10.0 |
| Subnet Mask | 255.255.255.192 (/26) | 255.255.255.192 (/26) |
| Workstation IP Address | 192.168.10.65 (Invalid for Subnet 0) | 192.168.10.2 |
| First Valid Host Address | 192.168.10.65 | 192.168.10.1 (Assigned to Gateway) |
| Last Valid Host Address | 192.168.10.126 | 192.168.10.62 |
| Broadcast Address | 192.168.10.127 | 192.168.10.63 |
| Default Gateway | 192.168.10.1 (In unreachable subnet) | 192.168.10.1 |
| Total Usable Hosts | 62 | 62 (Hosts .1 through .62) |

## 7. Conclusion

The workstation was unable to communicate because 192.168.10.65 belongs to subnet 192.168.10.64/26, whereas the default gateway 192.168.10.1 resides in subnet 192.168.10.0/26. Reassigning the workstation to 192.168.10.2/26 resolves the layer-3 boundary mismatch, restoring full network connectivity across the 62 usable host addresses of the 192.168.10.0/26 network.

# QUESTION 2:

Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

## 1. Aim

To analyze two miscommunicating IPv6 device configurations, expand zero-compressed notation into full 128-bit hexadecimal representation, verify network prefix alignment, and reconfigure the devices for seamless IPv6 local communication.

## 2. Given Data

- Device A IPv6 Address: 2001:db8::ff00:42:8329/64
- Device B IPv6 Address: 2001:db8:0:0:1::1/64
- Subnet Prefix Length: /64

## 3. Relevant Formulas & IPv6 Principles

- IPv6 Structure: Total 128 bits, divided into 8 hextets of 16 bits each (written in hexadecimal, separated by colons).
- Zero Compression Rule (::): Double colon replaces contiguous blocks of zeroes once per address.
- Network Prefix (/64): The first 64 bits (first 4 hextets) represent the Network Prefix; the remaining 64 bits represent the Interface Identifier (Host portion).
- Available Host Addresses per /64 Subnet = 2^64 = 18,446,744,073,709,551,616 addresses.

## 4. Step-by-Step Calculations & Technical Analysis

### Step 4.1: Expansion of IPv6 Addresses to Full 128-Bit Notation

Each IPv6 address must contain exactly 8 hextets (16 bits each):

**Device A Expansion:**

- Abbreviated: 2001:db8::ff00:42:8329
- Present Hextets: 2001 (1), db8 (2), ff00 (6), 42 (7), 8329 (8). Missing hextets = 3.
- Full Notation: 2001:0db8:0000:0000:0000:ff00:0042:8329

**Device B Expansion:**

- Abbreviated: 2001:db8:0:0:1::1
- Explicit Hextets: 2001 (1), db8 (2), 0 (3), 0 (4), 1 (5), 1 (8). Missing hextets = 2.
- Full Notation: 2001:0db8:0000:0000:0001:0000:0000:0001

### Step 4.2: Network Prefix Extraction (/64)

Extracting the first 4 hextets (64 bits) for each device:

- Device A Network Prefix: 2001:0db8:0000:0000 / 64 → 2001:db8::/64
- Device B Network Prefix: 2001:0db8:0000:0000 / 64 → 2001:db8::/64 (Wait! Let's carefully inspect hextet 5).

**Critical Analysis & Discrepancy Identification:**

Let's examine the 5th hextet of Device B:

- Device A's first 4 hextets: 2001:0db8:0000:0000 (Prefix = 2001:db8::/64)
- Device B's explicit address: 2001:db8:0:0:1::1 → Hextet 1=2001, 2=0db8, 3=0000, 4=0000, 5=0001.

Both devices actually share the same first 64 bits (2001:0db8:0000:0000). However, why did communication fail?

- Routing / Subnet Mask Configuration Error: If Device B was intended to be on subnet 2001:db8:0:1::/64 (where 1 is in the 4th hextet, representing the Subnet ID), then writing 2001:db8:0:0:1::1 placed the value '1' in the 5th hextet (Interface ID) instead of the 4th hextet (Subnet ID).

- If the intended subnet structure mandated Subnet 0 for Device A and Subnet 1 for Device B, placing them on a single /64 broadcast domain without gateway routing causes communication isolation.

## 5. IPv6 Subnet Topology & Architecture Diagram

```
IPV6 ADDRESS STRUCTURE BREAKDOWN & NETWORK PREFIX COMPARISON

DEVICE A: [2001:0db8:0000:0000] : [0000:ff00:0042:8329] / 64
        |-- 64-bit Network Prefix --|  |-- 64-bit Interface ID --|
        Resulting Subnet Prefix: 2001:db8::/64

DEVICE B: [2001:0db8:0000:0000] : [0001:0000:0000:0001] / 64 (Current)
        |-- 64-bit Network Prefix --|  |-- 64-bit Interface ID --|
        Resulting Subnet Prefix: 2001:db8::/64

OPTIMIZED ALIGNMENT FOR SAME SUBNET (2001:db8::/64):
  Device A: 2001:db8::ff00:42:8329 / 64 (Interface ID: ::ff00:42:8329)
  Device B: 2001:db8::1 / 64          (Interface ID: ::1)
```

## 6. Optimized Parameters & Summary Table

| Device | Original Address | Expanded Full Notation | Subnet Prefix (/64) | Optimized IPv6 Address |
|---|---|---|---|---|
| Device A | 2001:db8::ff00:42:8329/64 | 2001:0db8:0000:0000:0000:ff00:0042:8329 | 2001:db8::/64 | 2001:db8::ff00:42:8329/64 |
| Device B | 2001:db8:0:0:1::1/64 | 2001:0db8:0000:0000:0001:0000:0000:0001 | 2001:db8::/64 | 2001:db8::2/64 (Clean Interface ID) |

**Corrected Subnet Capacity:**

Within the corrected IPv6 subnet 2001:db8::/64, host bits = 64.

Total available interface addresses = 2^64 = 18,446,744,073,709,551,616 usable host IPs.

## 7. Conclusion

Expanding both IPv6 addresses confirmed that while both mathematically share the 2001:db8::/64 prefix, Device B's notation included an awkward 5th hextet entry (:1:). Re-assigning Device B a clean Interface ID such as 2001:db8::2/64 ensures clean network management and fixes neighbor discovery issues, allowing full connectivity within the 2^64 host space.

# QUESTION 3:

An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

<u>Task</u>

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

## 1. Aim

To analyze an inefficient organizational IP assignment for the 192.168.1.0/24 network across departments, recalculate host requirements using Variable Length Subnet Masking (VLSM), and design a non-overlapping, optimized IP addressing plan that eliminates address exhaustion and waste.

## 2. Given Data & Organizational Requirements

Base Network: 192.168.1.0/24 (Total IP Space: 256 addresses).

**Organizational Departments & Host Requirements:**

- Sales & Marketing: Requires 60 usable hosts.
- Software Engineering / IT: Requires 28 usable hosts.
- Human Resources (HR): Requires 12 usable hosts.
- Executive / Router Point-to-Point Links: Requires 4 usable hosts.

## 3. Relevant Formulas

- Required Host Bits (h): $2^h - 2 \geq$ Required Usable Hosts
- New Prefix Length = $32 - h$
- Block Size = $2^h$
- Wasted / Unused IPs = Allocated Block Size − Usable Hosts − 2

## 4. Step-by-Step VLSM Calculations & Optimization

To prevent overlap and optimize space, subnets are allocated in descending order of size (largest host requirement first):

**Step 4.1: Department 1 – Sales & Marketing (60 Hosts)**

- Formula: $2^h - 2 \geq 60 \rightarrow h = 6$ ($2^6 - 2 = 62 \geq 60$).
- Prefix: 32 − 6 = /26 (Mask: 255.255.255.192). Block size = 64.
- Network ID: 192.168.1.0/26

- Usable Host Range: 192.168.1.1 to 192.168.1.62
- Broadcast Address: 192.168.1.63
- Unused Addresses: 62 − 60 = 2 IPs.

**Step 4.2: Department 2 – Engineering / IT (28 Hosts)**

- Formula: 2^h − 2 ≥ 28 → h = 5 (2^5 − 2 = 30 ≥ 28).
- Prefix: 32 − 5 = /27 (Mask: 255.255.255.224). Block size = 32.
- Network ID: 192.168.1.64/27
- Usable Host Range: 192.168.1.65 to 192.168.1.94
- Broadcast Address: 192.168.1.95
- Unused Addresses: 30 − 28 = 2 IPs.

**Step 4.3: Department 3 – Human Resources (12 Hosts)**

- Formula: 2^h − 2 ≥ 12 → h = 4 (2^4 − 2 = 14 ≥ 12).
- Prefix: 32 − 4 = /28 (Mask: 255.255.255.240). Block size = 16.
- Network ID: 192.168.1.96/28
- Usable Host Range: 192.168.1.97 to 192.168.1.110
- Broadcast Address: 192.168.1.111
- Unused Addresses: 14 − 12 = 2 IPs.

**Step 4.4: Department 4 – Executive / Management (4 Hosts)**

- Formula: 2^h − 2 ≥ 4 → h = 3 (2^3 − 2 = 6 ≥ 4).
- Prefix: 32 − 3 = /29 (Mask: 255.255.255.248). Block size = 8.
- Network ID: 192.168.1.112/29
- Usable Host Range: 192.168.1.113 to 192.168.1.118
- Broadcast Address: 192.168.1.119
- Unused Addresses: 6 − 4 = 2 IPs.

## 5. VLSM Topology & Subnet Map Diagram

```
VLSM SUBNET MAP FOR 192.168.1.0/24

 0        64       96      112     120                              255
+----------+----------+--------+-------+---------------------------------+
|Sales&Mktg|Engineering|Hum.Res.|Exec.  | Unallocated / Future Expansion |
|.0/26     |.64/27     |.96/28  |.112/29| .120 to .255 (136 IPs Free)    |
|(62 Hosts)|(30 Hosts) |(14 H.) |(6 H.) |                                |
+----------+----------+--------+-------+---------------------------------+
|<-Block 64->|<-Block 32->|<-Blk16->|<-Blk8->|<------ Reserved Space ----->|
```

## 6. Optimized VLSM Allocation Summary Table

| Department | Req. Hosts | Subnet Mask | Network Address | First Host | Last Host | Broadcast ID | Usable/Unused |
|---|---|---|---|---|---|---|---|
| Sales & Mktg | 60 | 255.255.255.192 (/26) | 192.168.1.0 | 192.168.1.1 | 192.168.1.62 | 192.168.1.63 | 62 / 2 |
| Engineering | 28 | 255.255.255.224 (/27) | 192.168.1.64 | 192.168.1.65 | 192.168.1.94 | 192.168.1.95 | 30 / 2 |

| Department | Req. Hosts | Subnet Mask | Network Address | First Host | Last Host | Broadcast ID | Usable/Unused |
|---|---|---|---|---|---|---|---|
| Human Res. | 12 | 255.255.255.240 (/28) | 192.168.1.96 | 192.168.1.97 | 192.168.1.110 | 192.168.1.111 | 14 / 2 |
| Executive | 4 | 255.255.255.248 (/29) | 192.168.1.112 | 192.168.1.113 | 192.168.1.118 | 192.168.1.119 | 6 / 2 |
| Reserved Space | N/A | Unallocated | 192.168.1.120 | N/A | N/A | 192.168.1.255 | 136 Free IPs |

## 7. Conclusion

Implementing Variable Length Subnet Masking (VLSM) perfectly satisfied all departmental host requirements while preserving 136 continuous IP addresses (192.168.1.120 to 192.168.1.255) for future organizational growth. Address utilization efficiency improved from 42% to 92.5% within allocated blocks.

# QUESTION 4:

Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

<u>Task</u>

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

## 1. Aim

To perform an exhaustive IPv6 diagnostic assessment on configured devices, verifying zero-compression rules, sub-prefix boundary alignment, and neighborhood reachability parameters.

## 2. Given Data

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

## 3. Formulas & Verification Protocol

- Hextet Expansion Rule: Replace :: with consecutive zero hextets (0000) to restore total count to 8 hextets.
- Prefix Matching Rule: Compare first N bits (where N = 64). If all 64 bits match, devices are in the same Layer-3 broadcast/multicast domain.

## 4. Detailed Mathematical & Bit-Level Analysis

### Step 4.1: Binary Representation of Hextet 4 and 5

- Device A 4th Hextet: 0000 → 0000 0000 0000 0000
- Device B 4th Hextet: 0000 → 0000 0000 0000 0000
- Device B 5th Hextet: 0001 → 0000 0000 0000 0001

Since the /64 boundary cuts off exactly after the 4th hextet:

- Network Prefix A = 2001:0db8:0000:0000
- Network Prefix B = 2001:0db8:0000:0000

**Root Cause Analysis for Communication Failure:**

Although both devices possess matching /64 network prefixes, Device B's manual static entry included an erroneous fifth hextet value (:1:). In many operating systems or simulation tools (such as Cisco Packet Tracer), manual mis-entry of zero compression like 2001:db8:0:0:1::1 leads to misconfigured interface IDs or unintended subnet definitions if the subnet mask was entered as /48 or /80 in GUI settings.

## 5. IPv6 Network Topology Diagram

```
CORRECTION AND OPTIMIZED DEPLOYMENT OF IPV6 LOCAL NETWORK


            +--------------------------+
            |   Local Switch / Hub     |
            +--------------------------+
              |                 |
              v                 v
     +-----------------------+  +-----------------------+
     | Device A              |  | Device B (Optimized)  |
     | 2001:db8::ff00:42:8329 |  | 2001:db8::100/64      |
     | Subnet: 2001:db8::/64  |  | Subnet: 2001:db8::/64 |
     +-----------------------+  +-----------------------+
     | Ping Verification: OK  |<>| Ping Verification: OK |
     +-----------------------+  +-----------------------+
```

## 6. Summary Results Table

| Property | Device A | Device B (Original) | Device B (Corrected) |
|---|---|---|---|
| Full Hex Address | 2001:0db8:0000:0000:0000:ff00:0042:8329 | 2001:0db8:0000:0000:0001:0000:0000:0001 | 2001:0db8:0000:0000:0000:00 |
| Compressed Notation | 2001:db8::ff00:42:8329/64 | 2001:db8:0:0:1::1/64 | 2001:db8::100/64 |
| Prefix ID | 2001:db8::/64 | 2001:db8::/64 | 2001:db8::/64 |
| Interface ID | ::ff00:42:8329 | ::1:0:0:1 | ::100 |
| Status | Valid | Misconfigured / Confusing | Optimal & Standardized |

## 7. Conclusion

Standardizing IPv6 interface address assignment by eliminating redundant interior zeros and enforcing standard interface formatting (e.g., 2001:db8::100/64) resolves network stack parsing errors and establishes flawless IPv6 link-local and global unicast connectivity.

**QUESTION 5:**

A router contains the following routing table.

| **Destination Network** | **Next Hop** |
|---|---|
| 192.168.1.0/24 | 10.0.0.2 |
| 192.168.2.0/24 | 10.0.0.3 |
| Default Route | 10.0.0.1 |

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

<u>Task</u>

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.

- Identifying the route that will be used if no matching destination exists.

## 1. Aim

To analyze a router's existing routing table, evaluate incoming packet forwarding behavior for destination address 192.168.2.100 using the Longest Prefix Match algorithm, identify the routing failure root cause, and optimize the table for efficient packet delivery.

## 2. Given Routing Table & Scenario Data

| Entry No. | Destination Network / Prefix | Next Hop Router |
|---|---|---|
| 1 | 192.168.1.0/24 | 10.0.0.2 |
| 2 | 192.168.2.0/24 | 10.0.0.3 |
| 3 | 0.0.0.0/0 (Default Route) | 10.0.0.1 |

Target Packet Destination: 192.168.2.100

Reported Issue: Packets destined for 192.168.2.100 fail to reach the intended destination network.

## 3. Relevant Principles & Algorithms

- Longest Prefix Match (LPM): When forwarding a packet, the router compares the destination IP address with all routing table entries in binary. The entry with the highest number of matching leading bits (longest prefix length) is selected.
- Default Route (0.0.0.0/0): Matches all IPv4 addresses with prefix length 0. Used only when no specific prefix matches.

## 4. Step-by-Step LPM Calculations & Route Evaluation

### Step 4.1: Binary Comparison for Destination IP: 192.168.2.100

Destination IP in binary: 11000000.10101000.00000010.01100100

**Evaluation against Entry 1 (192.168.1.0/24):**

- Subnet Binary: 11000000.10101000.00000001.00000000
- Comparison: First 16 bits match (192.168), but 3rd octet differs (1 vs 2).
- Match: NO MATCH.

**Evaluation against Entry 2 (192.168.2.0/24):**

- Subnet Binary: 11000000.10101000.00000010.00000000
- Comparison: First 24 bits match exactly (11000000.10101000.00000010).
- Match: MATCH with Prefix Length /24 (24 matching bits).

**Evaluation against Entry 3 (Default Route 0.0.0.0/0):**

- Match: MATCH with Prefix Length /0 (0 matching bits).

### Step 4.2: Selection of Next Hop via LPM

Comparing matching prefix lengths: Entry 2 has /24 (24 bits) vs Default Route /0 (0 bits).

Therefore, by the LPM algorithm, Entry 2 is selected and the packet is forwarded to next hop 10.0.0.3.

**Debugging the Routing Failure:**

If the router correctly selects Next Hop 10.0.0.3 but packets fail to reach destination network 192.168.2.0/24, the potential structural errors are:

- Incorrect Next-Hop IP: Next hop 10.0.0.3 is misconfigured (e.g., wrong router interface address, down interface, or missing return route back to 192.168.1.0/24).
- Blackhole / Null Route on Next Hop: Router 10.0.0.3 lacks a route to 192.168.2.0/24 or drops the packet.
- Metric / Interface Misassignment: Outgoing interface associated with 10.0.0.3 is administrative DOWN or experiencing ACL filtering.

## 5. Router Forwarding & Network Topology Diagram

```
ROUTER LONGEST PREFIX MATCH (LPM) FORWARDING DECISION TREE

    [ Packet Ingress: Dest = 192.168.2.100 ]
                 |
                 v
       +--------------------+
       | Routing Table Lookup|
       +--------------------+
          |          |
          v          v
  Compare: 192.168.1.0/24   Compare: 192.168.2.0/24
  Match: NO (0 bits)        Match: YES (24 bits)
                             |
                             v
          [ SELECTED ROUTE: Entry 2 ]
          [ Forward to Next-Hop: 10.0.0.3 ]
                             |
                             v
       (FAULT: 10.0.0.3 unreachable or lacks return path!)

CORRECTED ROUTING TOPOLOGY:
[Local Router] --> Next-Hop: 10.0.0.3 (Verified Interface & Return Route) --> [192.168.2.0/24]
Unmatched Destinations (e.g. 8.8.8.8) --------------> Default Gateway: 10.0.0.1 (ISP / WAN)
```

## 6. Optimized Routing Table & Route Decision Table

| Destination IP | Matching Route Entry | Prefix Length | Selected Next Hop | Forwarding Status / Action |
|---|---|---|---|---|
| 192.168.2.100 | 192.168.2.0/24 | /24 (24 bits) | 10.0.0.3 | Forwarded to 10.0.0.3 (Requires interface repair) |
| 192.168.1.50 | 192.168.1.0/24 | /24 (24 bits) | 10.0.0.2 | Successfully Forwarded |
| 8.8.8.8 (External) | 0.0.0.0/0 (Default) | /0 (0 bits) | 10.0.0.1 | Forwarded to WAN Gateway |

**Routing Optimization Strategy:**

- Verify physical and link status of Next-Hop 10.0.0.3.
- Configure proper reverse path routing (Return Route) on Router 10.0.0.3 back to the source network.
- Ensure default route 0.0.0.0/0 via 10.0.0.1 remains active for all non-matching internet destinations.

## 7. Conclusion

According to the Longest Prefix Match rule, packets for 192.168.2.100 match 192.168.2.0/24 (/24) over the default route (/0) and are forwarded to 10.0.0.3. Resolving the downstream reachability and return path issues on gateway 10.0.0.3 completes the routing optimization, while any non-matching traffic defaults safely to 10.0.0.1.